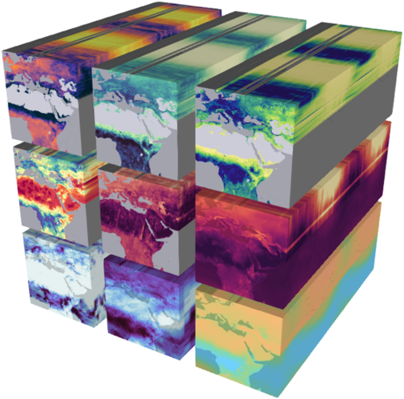
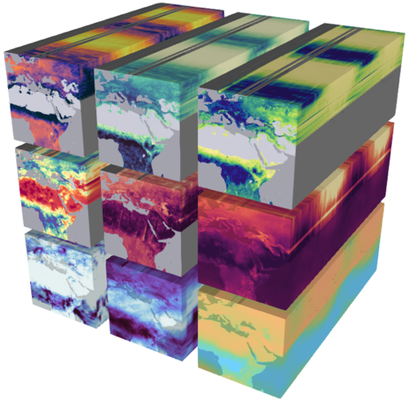
A Friendly Introduction to Dimensionality Reduction

Nathan Mankovich

Earth systems data



Earth system data cubes Mahecha et al. [2020].



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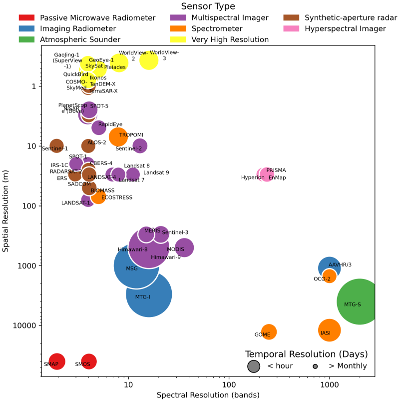
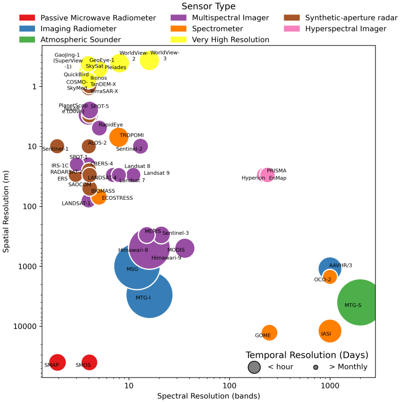


Dimensionality of RS data modalities.

Earth-systems data is high dimensional



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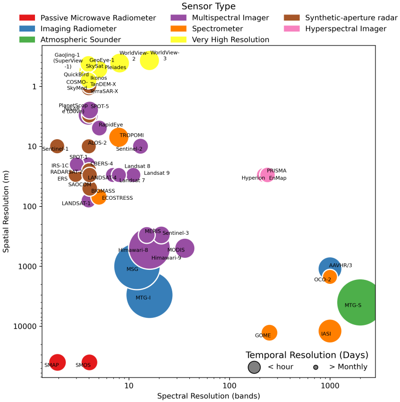

- Feature dimensions: spatial, spectral, temporal
- Sensors are not often high-dimensional in all dimensions
- Different data modalities in RS & outside of RS

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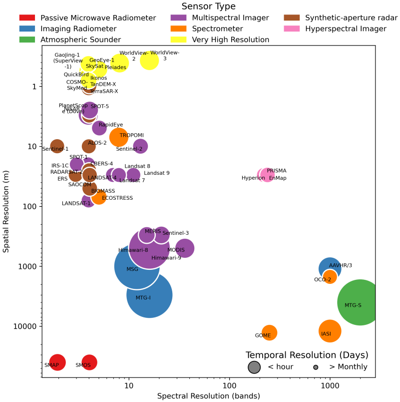

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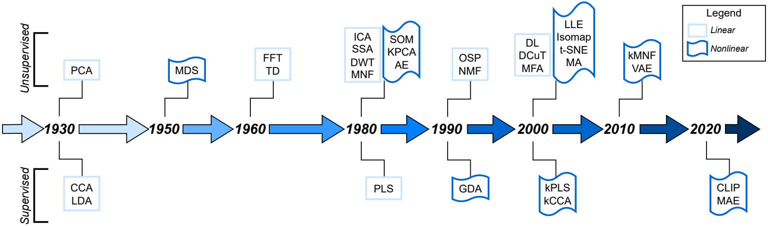
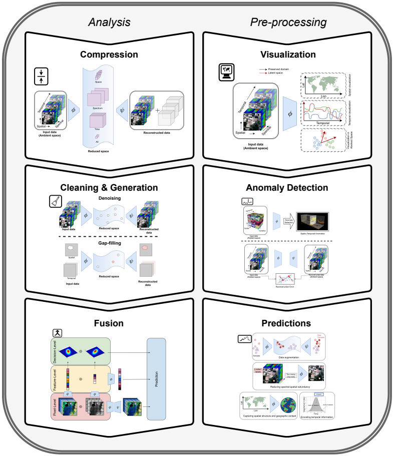
Dimensionality Reduction (DR) extracts a smaller number of discriminatory features.

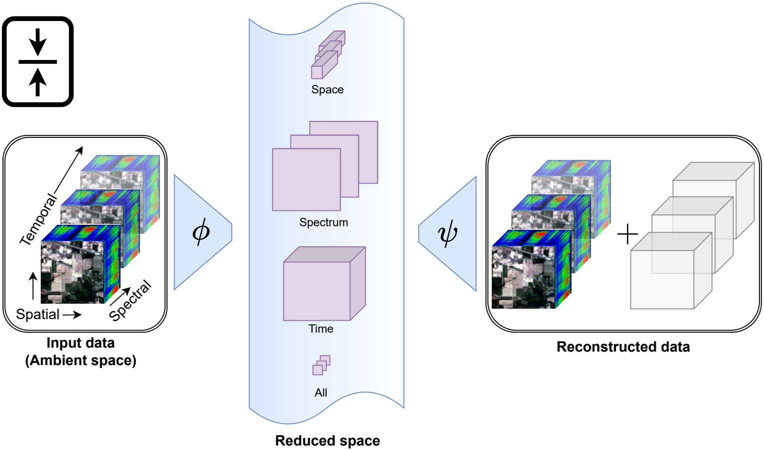
Dimensionality Reduction over the Years



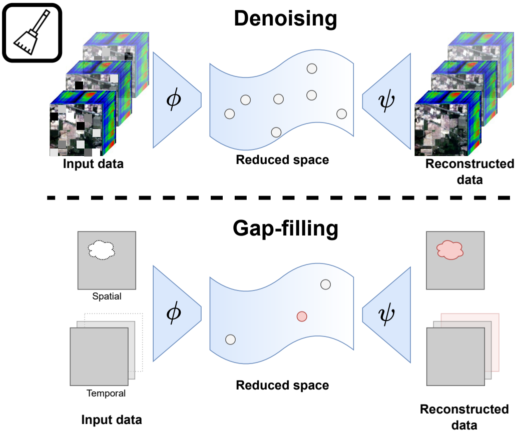
Dimensionality Reduction in Remote Sensing



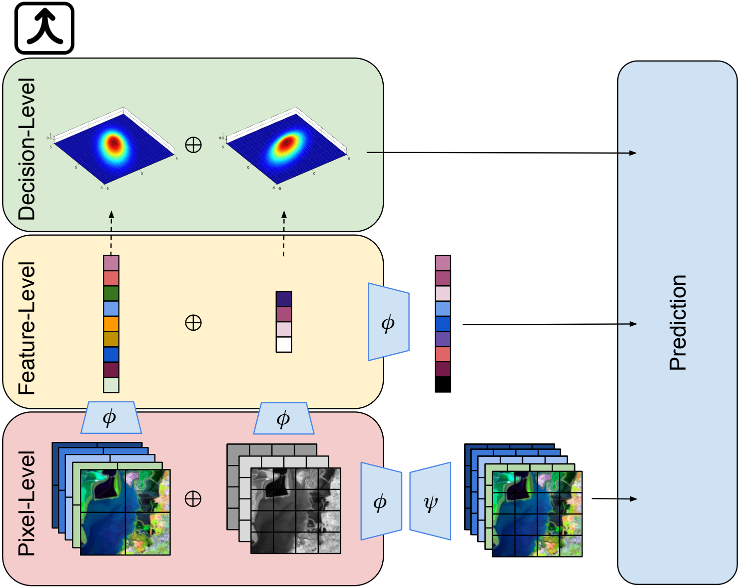
Compression



Denoising

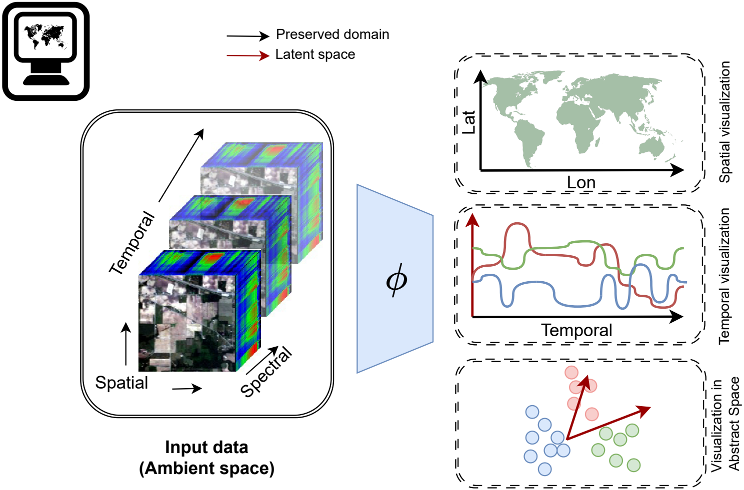

Denoising
Input data
ϕ
Reduced space
ψ
Reconstructed data
Gap-filling
Spatial
Temporal
Input data
ϕ
Reduced space
ψ
Reconstructed data

Fusion



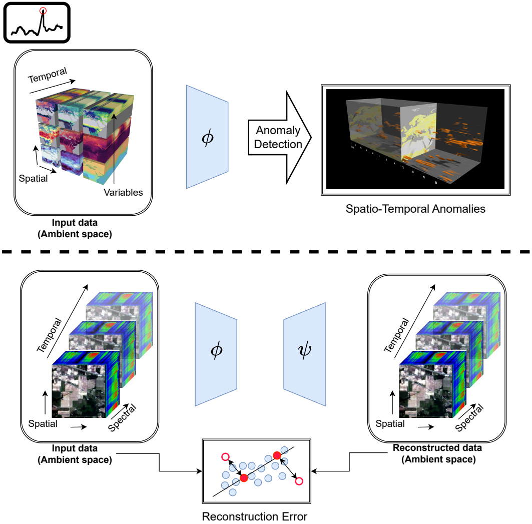
Decision-Level
Feature-Level
Pixel-Level
Prediction
$\oplus$
ϕ
ψ

Visualization

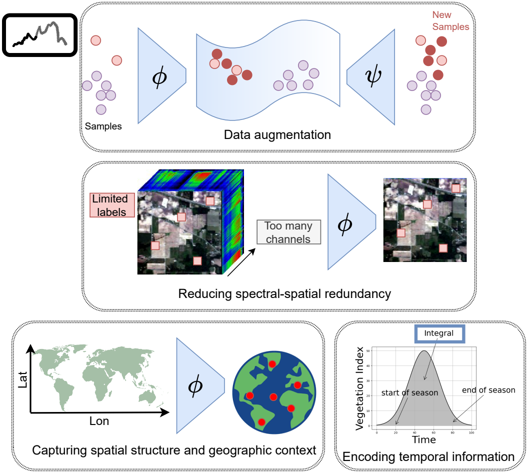


Input data
(Ambient space)

Anomaly Detection



Improved Predictions



Dimensionality Reduction in Remote Sensing (Full Article)

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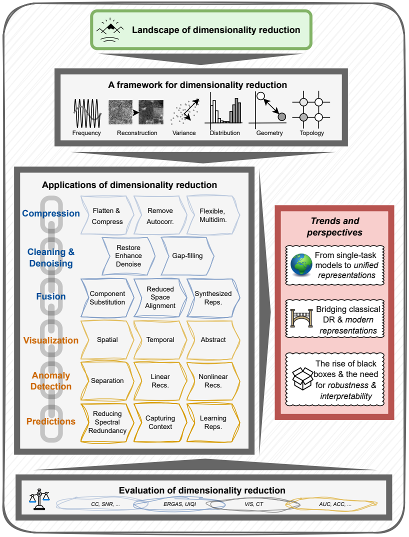


Dimensionality Reduction for Remote Sensing Data Analysis: A Systematic Review of Methods and Applications

Nathan Mankovich, Kai-Hendrik Cohrs, Homer Durand, Vasileios Sitokonstantinou, Tristan Williams, and Gustau Camps-Valls

Image Processing Lab, Universitat de València, C/ Cat. Agustín Escardino Benlloch, 9, Paterna, 46980, València, Spain

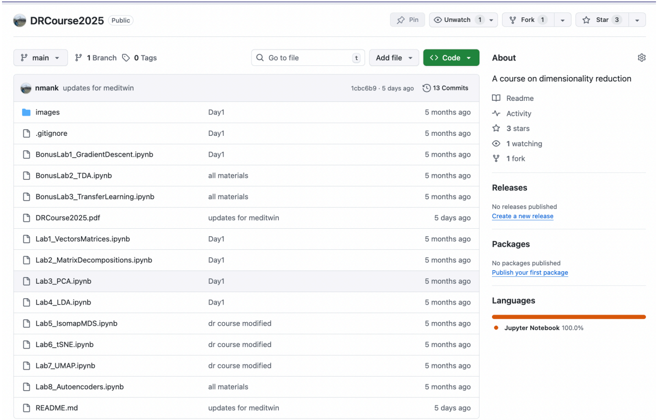
Link to article:
https://arxiv.org/pdf/2510.18935

Course Outline

1. Linear Algebra Review
2. Intro to DR
3. Linear Dimensionality Reduction
4. Nonlinear Dimensionality Reduction
5. Autoencoders

Course Webpage

https://github.com/nmank/DRCourse2025



Linear Algebra Review

Further reading [Strang, 2000]

Outline

1. Vectors & Inner Products
2. Subspaces, Orthogonality, Projections
3. Matrix Decompositions
4. Gradient Descent

Vectors & Inner Products

Field Axioms

A field F is a set equipped with two operations (addition and multiplication) satisfying the following axioms for all $a, b, c \in F$:

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- **Inverses:**
  - Additive inverse: for every $a \in F$, there exists $-a \in F$ such that $a + (-a) = 0$
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- **Distributivity:** $a \cdot (b + c) = (a \cdot b) + (a \cdot c)$

The set of all real numbers (denoted $\mathbb{R}$) is a Field

- Addition and multiplication operations
- Additive identity is 0
- Multiplicative identity is 1

Exercise: Show that $\mathbb{R}$ satisfies the axioms of a field, whereas $\mathbb{Z}$ (the set of all integers) does not.

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- **Commutativity of addition:** $\mathbf{u} + \mathbf{v} = \mathbf{v} + \mathbf{u}$
- **Additive identity:** There exists $\mathbf{0} \in V$ such that $\mathbf{v} + \mathbf{0} = \mathbf{v}$
- **Additive inverse:** For each $\mathbf{v} \in V$, there exists $-\mathbf{v} \in V$ such that $\mathbf{v} + (-\mathbf{v}) = \mathbf{0}$
- **Compatibility with scalar multiplication:** $a(b\mathbf{v}) = (ab)\mathbf{v}$
- **Identity element of scalar multiplication:** $1\mathbf{v} = \mathbf{v}$, where 1 is the multiplicative identity in F
- **Distributivity over vector addition:** $a(\mathbf{u} + \mathbf{v}) = a\mathbf{u} + a\mathbf{v}$
- **Distributivity over field addition:** $(a + b)\mathbf{v} = a\mathbf{v} + b\mathbf{v}$

$\mathbb{R}^2$ is a Vector Space

$\mathbb{R}^2$ is a Vector Space over $\mathbb{R}$.

$$\mathbb{R}^2 = \left\{ \mathbf{a} = \begin{bmatrix} a_1 \\ a_2 \end{bmatrix} \ : \ a_1, a_2 \in \mathbb{R} \right\}$$

Exercise: Show that $\mathbb{R}^2$ satisfies the axioms of a vector space.

Dot Product & Friends

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- **Angle between vectors**

$$\cos(\theta) = \frac{\mathbf{a}^\top \mathbf{b}}{\|\mathbf{a}\|_2 \|\mathbf{b}\|_2}$$

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$$\text{span}(\mathbf{a}, \mathbf{b}) = \alpha\mathbf{a} + \beta\mathbf{b} \ : \ \alpha, \beta \in \mathbb{R}$$

Linear Independence

A set of vectors $\{v_1, v_2, \ldots, v_k\}$ in a vector space V is **linearly independent** if:

$$c_1 v_1 + c_2 v_2 + \cdots + c_k v_k = 0 \quad \Rightarrow \quad c_1 = c_2 = \cdots = c_k = 0$$

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Example:

$$\begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad \begin{bmatrix} 0 \\ 1 \end{bmatrix}$$

are linearly independent in $\mathbb{R}^2$, since neither can be written as a multiple of the other.

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Example: The standard basis for $\mathbb{R}^3$ is:

$$\left\{ \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ 0 \\ 1 \end{bmatrix} \right\}$$

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The **dimension** of a vector space V is the number of elements in a basis for V.

Subspaces, Orthogonality, Projections

Matrices

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$$

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- The **rank** of **A** is the dimension of its column space.
- The **trace** of **A** is the sum of the diagonal entries

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- The Frobenius norm of $\mathbf{A}$ is

$$\|\mathbf{A}\|_F = \sqrt{\mathrm{tr}(\mathbf{A}^\top \mathbf{A})}$$

Matrix multiplication

$$\begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} \\ b_{21} & b_{22} \end{bmatrix}$$

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Projections

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Projection of **b** onto col(**A**):

$$\mathbf{\Pi}_{\mathbf{A}}(\mathbf{b}) = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1}\mathbf{A}^\top \mathbf{b}$$

Projections

Projection of $\mathbf{b}$ onto col($\mathbf{A}$):

$$\mathbf{\Pi}_{\mathbf{A}}(\mathbf{b}) = \mathbf{A}(\mathbf{A}^\top \mathbf{A})^{-1}\mathbf{A}^\top \mathbf{b}$$

Exercise: If the columns of $\mathbf{A}$ are orthogonal, show that the projection onto the column space of $\mathbf{A}$ is $\mathbf{A}\mathbf{A}^\top$.

Lab 1: Vectors & Matrices

Go to Lab1_VectorsMatrices.ipynb

Matrix Decompositions

Singular Value Decomposition (SVD)

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- $\boldsymbol{\Sigma} \in \mathbb{R}^{r \times r}$: diagonal matrix of singular values
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Applications: dimensionality reduction, image compression, linear systems.

Eigenvalue Decomposition from SVD

Given the SVD

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- **Eigenvectors**: columns of $\mathbf{V}$
- **Eigenvalues**: diagonal of $\boldsymbol{\Lambda}$

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- **Eigenvectors**: columns of $\mathbf{V}$
- **Eigenvalues**: diagonal of $\boldsymbol{\Lambda}$

Exercises:

- Show the eigenvalue decomposition of $\mathbf{A}\mathbf{A}^\top$ is $\mathbf{U}\boldsymbol{\Sigma}\boldsymbol{\Sigma}^\top\mathbf{U}^\top$
- Show the trace of $\mathbf{C}$ is the sum of its eigenvalues

Eigenvalue Optimization Formulation

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1. **Rayleigh quotient:**

$$\lambda_i = \max_{\mathbf{v}^\top \mathbf{v}_j = 0 \; \forall j < i} \frac{\mathbf{v}^\top \mathbf{C} \mathbf{v}}{\mathbf{v}^\top \mathbf{v}}$$

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Exercise: Show that these optimizations are equivalent.

Generalized Eigenvalue Decomposition

Solve:

$$\mathbf{C}_A\mathbf{w} = \lambda \mathbf{C}_B\mathbf{w}$$

Optimization form:

$$\lambda_i = \max_{\mathbf{w}^\top \mathbf{w}_j = 0\ \forall j<i} \frac{\mathbf{w}^\top \mathbf{C}_A \mathbf{w}}{\mathbf{w}^\top \mathbf{C}_B \mathbf{w}}$$

Approximate with:

$$\mathbf{C}_B^{-1}\mathbf{C}_A$$

Use SVD for pseudo-inverse:

- $\mathbf{C}_B = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\top$
- $\mathbf{C}_B^\dagger = \mathbf{V}\boldsymbol{\Sigma}^{-1}\mathbf{U}^\top$

Condition number:

$$\kappa(\mathbf{C}_B) = \frac{\sigma_{\max}}{\sigma_{\min}}$$

Big κ bad, Small κ good

Lab 2: Matrix Decompsitions

Go to Lab2_MatrixDecompositions.ipynb

The Landscape of Dimensionality Reduction

Inspired by Lee and Verleysen [2007]

DR Reduces the Feature Space

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Dimensionality Reduction (DR) maps d-dimensional features to $k < d$-dimensional "essential" features

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Many DR algorithms & definitions of "essential"

Toy Example: Circles

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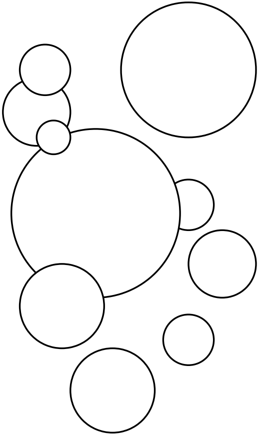
Ambient space



Reduced space

(0, 0), 1

(2, 3), 2

(-1, 4), 1.5

(3, -2), 0.8

(-2, -3), 2.5

(1, -1), 1.2

(-3, 1), 0.9

(4, 4), 1.8

(-4, -2), 1.1

(0, 5), 2

DR Summary- Properties

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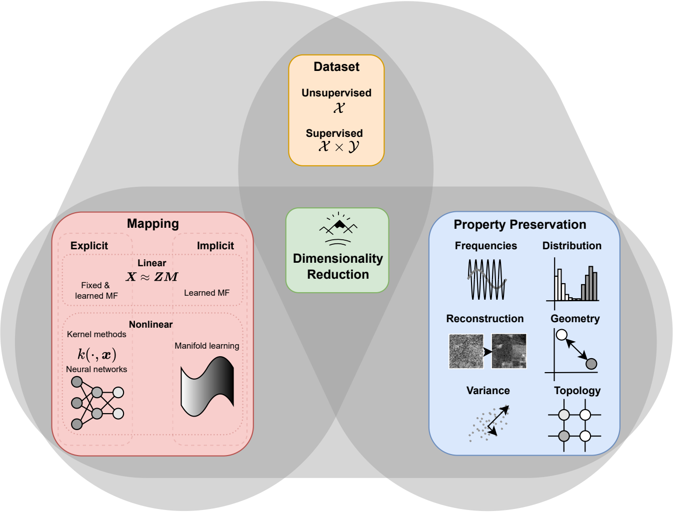

Dataset

Unsupervised
$\mathcal{D} = \mathcal{X} = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_p\} \subset \mathbb{R}^d$

e.g., Visualize data in 2D to see if there are any patterns

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Supervised
$\mathcal{D} = \{\mathcal{X}, \mathcal{Y}\}, = \{\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_p\}, \{\mathbf{y}_1, \mathbf{y}_2, \ldots, \mathbf{y}_p\}$

e.g., Find a low-dimensional, discriminatory feature space w.r.t. $\mathcal{Y}$

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Semi-supervised & self-supervised methods break this paradigm... but we will not discuss these methods in this course

DR Mapping

Explicit DR mapping outputs ϕ where

$$\phi(\mathbf{x}_i) = \mathbf{z}_i$$

- Approximate inverse $\psi \approx \phi^{-1}$
- Reconstructions: $\mathbf{x}_i \approx \hat{\mathbf{x}}_i = \psi(\phi(\mathbf{x}_i))$

e.g., Want to fit a model on some data, then apply it to "unseen" data.

Implicit: ϕ, ψ are not explicitly defined but inferred

e.g., Model fit on all the data to be reduced

Matrix Factorization

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Data matrix: $\mathbf{X} = [\mathbf{x}_1 \cdots \mathbf{x}_p]^\top \in \mathbb{R}^{p \times d}$

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The model:

$$\mathbf{X} \approx \mathbf{Z}\boldsymbol{\Psi}$$

Example: principal component analysis factors data matrix **X** into

- Principal components (rows of **Z**)
- Empirical orthogonal functions (columns of $\boldsymbol{\Psi}$)

Kernel Methods: Nonlinear Generalizations of Matrix Factorization

- Classical DR: **explicit linear mapping**

$$\mathbf{z} = \mathbf{\Psi}^\top \mathbf{x}$$

- Kernel methods extend classical DR to **nonlinear setting** by mapping into a high-dimensional feature space

$$\phi : \mathbf{x} \mapsto \mathcal{H},$$

  where linear DR is performed.

- **Kernel trick** replaces explicit feature mappings with pairwise similarities

$$k(\mathbf{x}_i, \mathbf{x}_j) = \langle \phi(\mathbf{x}_i), \phi(\mathbf{x}_j) \rangle,$$

  enabling nonlinear embeddings without constructing ϕ.

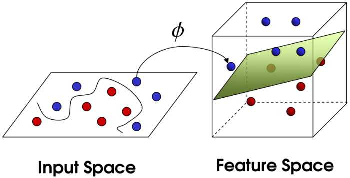

Figure: Image from https://medium.com/data-science/the-kernel-trick-c98cdbcaeb3f

Neural Networks

- Neural network–based DR methods learn a **parametric, nonlinear mapping**

$$f_\theta : \mathbf{x} \mapsto \mathbf{z},$$

- f_θ defined by a composition of affine transformations and nonlinear activations.
- Like kernel methods, mapping is learned **explicitly**.
- Objective function defines DR properties (discussed later)

Example Autoencoder (AE)

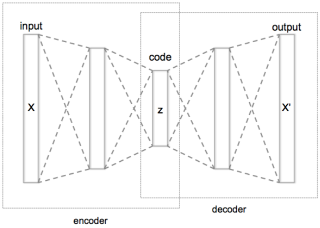

Figure: Image from `https://commons.wikimedia.org/wiki/File:Autoencoder_structure.png`

- Encoder: $\phi(\mathbf{x}) = \mathbf{z}$,
  Decoder: $\psi(\mathbf{z}) = \hat{\mathbf{x}}$
- Loss: e.g., MSE $\sum_i \|\mathbf{x}_i - \psi \circ \phi(\mathbf{x}_i)\|^2$

Manifold Learning: Implicit Mapping via Data Geometry

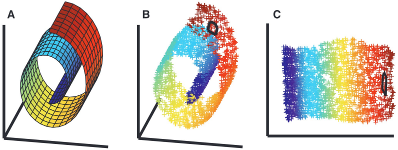

Figure: Image from https://www.dissonances.blog/p/ambient-spaces.

- Assume high-dimensional observations lie on or near a low-dimensional **manifold** embedded in the ambient space.
- **Implicit** mappings to low-dimensional representations by preserving local geometric or topological relationships.
- The embedding is typically obtained by solving a graph-based or spectral problem derived from neighborhood relations in the data.
- The resulting coordinates describe the intrinsic manifold structure.

DR as Optimization

DR reduced space representation $\mathcal{Z}$ is found as minimizer of a loss function L of the data $\mathcal{X}$.

$$\begin{gathered}\underset{\mathcal{Z}}{\arg\min} \quad L(\mathcal{X}, \mathcal{Z}) \\ \text{s.t. } \Omega(\mathcal{Z}, \mathcal{X}) = 0\end{gathered}$$

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but not all properties can be preserved at once

Frequency and signal structure preserving

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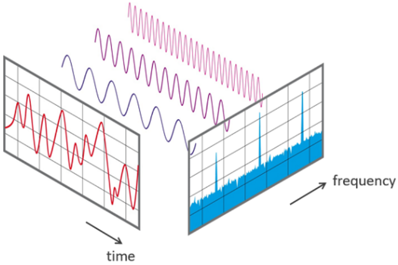

Figure: Image from https://www.nti-audio.com/en/support/know-how/fast-fourier-transform-fft

Reconstruction-preserving

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For each $i = 1, 2, \ldots, p$ we have

$$\mathbf{x}_i \approx \psi \circ \phi(\mathbf{x}_i)$$

Loss function

$$\begin{aligned} &\min_{\psi,\phi} \sum_{i=1}^{p} \|\mathbf{x}_i - \psi \circ \phi(\mathbf{x}_i)\|^2 \\ &\text{s.t. } \phi : \mathbb{R}^d \to \mathbb{R}^k, \quad \psi : \mathbb{R}^k \to \mathbb{R}^d \end{aligned} \tag{1}$$

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Useful when interpretability or invertibility is important. Tasks like compression, denoising, inpainting, generative modeling.

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Data $\mathbf{x}$ and DR mapping $\phi : \mathbb{R}^d \to \mathbb{R}^k$

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Interpretable and useful for detecting modes or variation, noise reduction, compression, and exploratory analysis.

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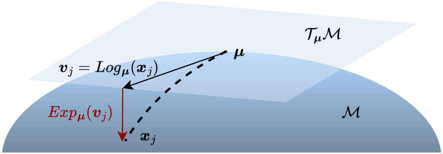

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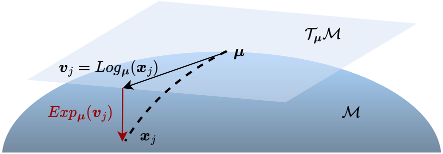

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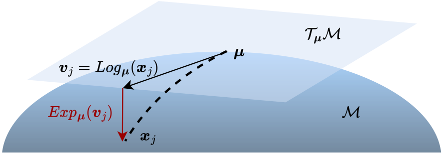

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Geodesic Distances: For $\mathbf{x}_i, \mathbf{x}_j \in \mathcal{M}$, define

$$d_{\mathcal{M}}(\mathbf{x}_i, \mathbf{x}_j) = \text{length of shortest path on } \mathcal{M}$$

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- Some approaches work directly from distance matrices rather than raw feature vectors.

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Data-driven topology (learned):

- Construct graph $G = (\mathcal{X}, E)$ from data.
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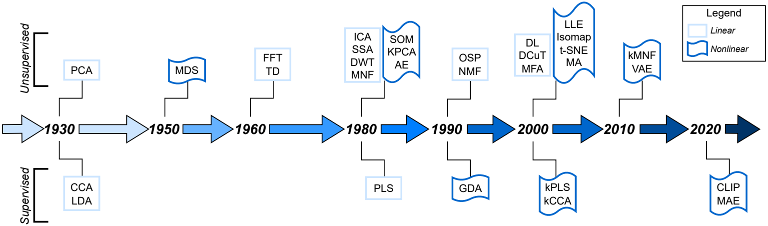
Predefined topology (fixed):

- Impose grid or lattice structure: $\mathcal{G} = 1\text{D}$ or 2D lattice.
- Each reduce-space representation $\mathbf{z}_i$ corresponds to a fixed node in $\mathcal{G}$.

Example: ISOMAP & t-distributed stochastic neighborhood embedding

Comparison:
Geometry-preserving $\Rightarrow$ maintain distances;
Topology-preserving $\Rightarrow$ maintain who is connected to whom.

DR Summary- Timeline



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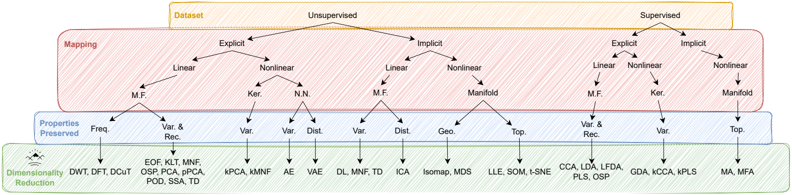

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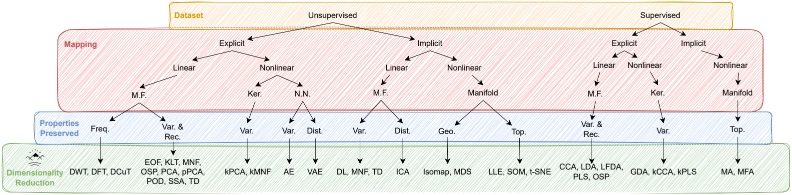

Useful git repository with a collection of many, many methods: `https://github.com/koulakis/awesome-dimensionality-reduction`

Linear Dimensionality Reduction

Outline

1. Principal Component Analysis [Hotelling, 1933; Shlens, 2014]

2. Linear Discriminant Analysis [Tharwat et al., 2017]

3. Dynamic Mode Decomposition [Schmid, 2010]

Principal Component Analysis

Dataset

We consider a dataset of p samples with d features:

$$\{\mathbf{x}_i\}_{i=1}^{p} \subset \mathbb{R}^d$$

We collect the dataset into a samples $\times$ features data matrix:

$$\mathbf{X} \in \mathbb{R}^{p \times d}$$

Important: The data must be **mean-centered**:

$$\frac{1}{p} \sum_{i=1}^{p} \mathbf{x}_i = \mathbf{0}$$

PCA Objective Summary

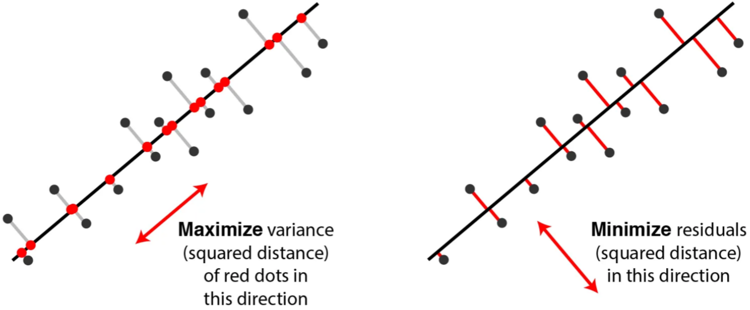

Figure: Image borrowed from `https://medium.com/@fraidoonomarzai99/principal-component-analysis-pca-in-depth-93c871f25dfa`.

Optimization Goal

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$$\mathbf{v}_j = \underset{\substack{\mathbf{v}^\top \mathbf{v}=1 \\ \mathbf{v}^\top \mathbf{v}_\ell=0\ \forall \ell<i}}{\arg\max}\ \mathrm{Var}\big(\mathbf{X}\mathbf{w}\big) \tag{3}$$

$$= \underset{\substack{\mathbf{w}^\top \mathbf{w}=1 \\ \mathbf{w}^\top \mathbf{v}_\ell=0\ \forall \ell<i}}{\arg\max}\ \sum_i (\mathbf{x}_i^\top \mathbf{w})^\top (\mathbf{x}_i^\top \mathbf{w}) \tag{4}$$

$$= \underset{\substack{\mathbf{w}^\top \mathbf{w}=1 \\ \mathbf{w}^\top \mathbf{v}_\ell=0\ \forall \ell<i}}{\arg\max}\ \sum_i \|\mathbf{x}_i\|_2^2 \cos\theta(\mathbf{w}, \mathbf{x}_i) \tag{5}$$

$$= \underset{\substack{\mathbf{w}^\top \mathbf{w}=1 \\ \mathbf{w}^\top \mathbf{v}_\ell=0\ \forall \ell<i}}{\arg\max}\ \mathbf{w}^\top \mathbf{X}^\top \mathbf{X}\mathbf{w} \tag{6}$$

Alternative Formulation: Reconstruction Error

This is equivalent to finding the rank-k projection:

$$\mathbf{\Pi}_{\mathbf{W}}(\mathbf{x}) := \mathbf{W}\mathbf{W}^\top \mathbf{x}$$

We solve:

$$\mathbf{V} = \arg \min_{\mathbf{W}^\top \mathbf{W} = \mathbf{I}} \mathbb{E}\left[\|\mathbf{x}_i - \mathbf{W}\mathbf{W}^\top \mathbf{x}_i\|_2^2\right]$$

Exercise Show that we can find the EOFs (columns of $\mathbf{V}$) via eigendecomposition of $\mathbf{X}^\top \mathbf{X}$ or the SVD of $\mathbf{X}$.

Transformation

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$$\hat{\mathbf{x}} = \psi \circ \phi(\mathbf{x}) = \mathbf{V}\mathbf{V}^\top \mathbf{x} = \mathbf{\Pi}_{\mathbf{V}}(\mathbf{x}) \in \mathbb{R}^d$$

Explained Variance

The **explained variance** is how much variance each principal component captures:

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Rule of thumb is select components that explain $> 90\%$ of variance. (but there's other methods [Gavish and Donoho, 2014])

PCA Summary

PCA Summary

Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.	Sup.	M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn

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When to use PCA

- Have unlabeled data

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Beyond PCA... robust subspace recovery [Lerman and Maunu, 2018] & flag manifolds [Mankovich et al., 2024; Szwagier and Pennec, 2024, 2025]

Aside- PCA on Manifolds (Tangent-PCA)

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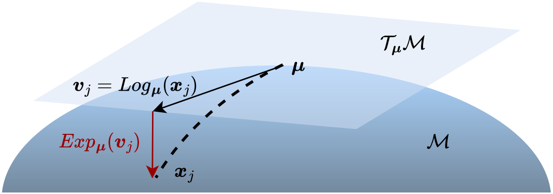

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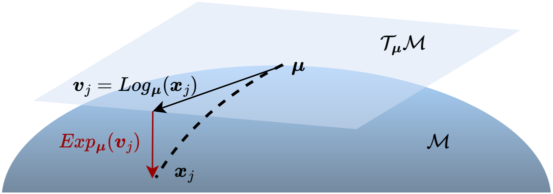

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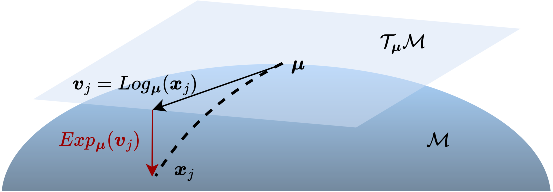

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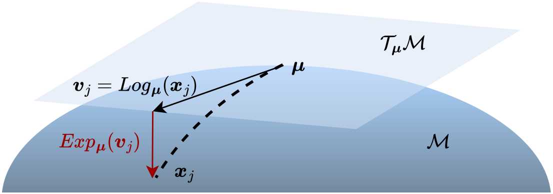

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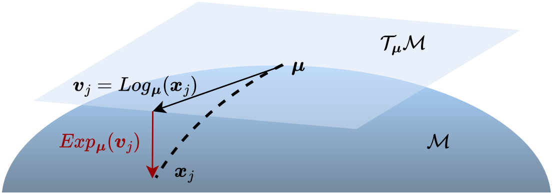

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Read more here: [Fletcher et al., 2004; Pennec, 2018]

Lab 3: PCA

Go to Lab3_PCA.ipynb

Linear Discriminant Analysis

What is LDA?

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"PCA with class information!"

LDA Intuition

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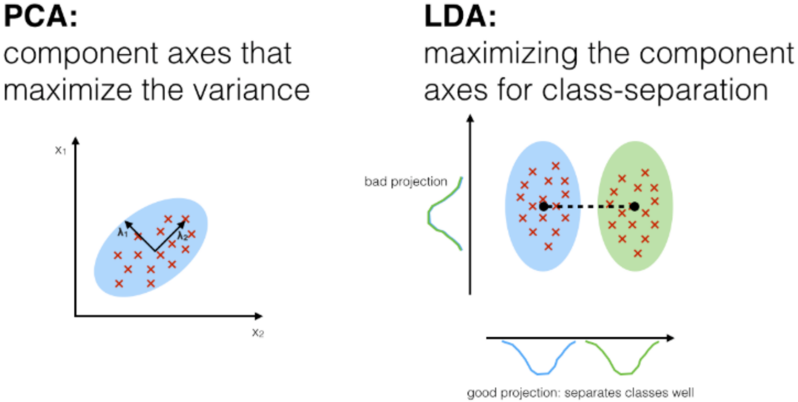

Figure: Image borrowed from
`https://sebastianraschka.com/Articles/2014_python_lda.html`

Mathematical Formulation

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Given labeled dataset:

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$$\mathcal{D} = \{(\mathbf{x}_i, y_i)\}_{i=1}^{p}, \quad \mathbf{x}_i \in \mathbb{R}^d$$

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Within-class scatter matrix:

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LDA vs PCA

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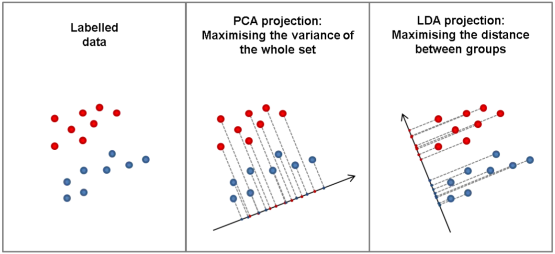

Figure: Image borrowed from `https://vivekmuraleedharan73.medium.com/what-is-linear-discriminant-analysis-lda-7e33ff59020a`.

Aspect	**PCA** [Shlens, 2014]	**LDA** [Tharwat et al., 2017]
Supervised?	No	Yes
Objective	Maximize variance	Maximize class separation
Axes chosen	max variance	best separation
Max dimensions	$\leq$ input dimension	$\leq$ number of classes -1

LDA Summary

Method	Map		Data	Constr. & Prop. Pres.							Code
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When to Use LDA

- Labeled data
- Want low-dimensional features that separate classes well

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When to Use LDA

- Labeled data
- Want low-dimensional features that separate classes well
- Want a fast, interpretable linear projection
- Linearly separable classes
- Each class follows a multivariate normal distribution with equal covariances

Lab 3: LDA

Go to Lab4_LDA.ipynb

Nonlinear Dimensionality Reduction

Outline

1. Multidimensional Scaling (MDS) [Kruskal and Wish, 1978]
2. Isometric Mapping (Isomap) [Balasubramanian and Schwartz, 2002]
3. t-distributed Stochastic Neighborhood Embedding (t-SNE) [Van der Maaten and Hinton, 2008]
4. Uniform Manifold Approximation (UMAP) [McInnes et al., 2018]

Multidimensional Scaling

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Given:

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- Distance matrix $\mathbf{D} \in \mathbb{R}^{p \times p}$, $d_{ij} = d(x_i, x_j)$

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Goal: Find latent coordinates $\{\mathbf{z}_i\}_{i=1}^p \subset \mathbb{R}^k$ such that

$$\|\mathbf{z}_i - \mathbf{z}_j\|_2 \approx d(x_i, x_j)$$

Useful when there's a non-Euclidean distance between points in the ambient space.

Classical MDS Objective

We solve:

$$\{\mathbf{z}_i\}_{i=1}^{p} = \arg \min_{\{\mathbf{w}_i\} \subset \mathbb{R}^k} \sum_{i,j} \left(\|\mathbf{w}_i - \mathbf{w}_j\|_2 - d_{ij} \right)^2$$

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Issue:

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Fix: Require mean centering:

$$\sum_{i=1}^{p} \mathbf{z}_i = \mathbf{0}$$

How do we mean-center a distance matrix?

Eigenvalue Decomposition

Let $\mathbf{D}^{(2)}$ be the matrix of squared distances: d_{ij}^2.

Mean-centering matrix:

$$\mathbf{J} = \mathbf{I} - \frac{1}{p}\mathbf{1}\mathbf{1}^\top$$

Double-mean center matrix of squared distances:

$$\mathbf{B} = -\frac{1}{2}\mathbf{J}\mathbf{D}^{(2)}\mathbf{J}$$

Compute eigenvalue decomposition:

$$\mathbf{B}\mathbf{V} = \mathbf{V}\boldsymbol{\Lambda}$$

Low-Dimensional Embedding

Truncate to top-k eigenvalues and eigenvectors:

$$\overline{\mathbf{\Lambda}},\ \overline{\mathbf{V}}$$

Construct the embedding:

$$\mathbf{Z} = \overline{\mathbf{V}}\,\overline{\mathbf{\Lambda}}^{1/2}$$

$\mathbf{Z}^\top = [\mathbf{z}_1, \dots, \mathbf{z}_n]$ are the low-dimensional coordinates.

Note: When $\mathcal{M} = \mathbb{R}^d$ and $d(x_i, x_j) = \|\mathbf{x}_i - \mathbf{x}_j\|_2$, MDS recovers PCA.
(Details: `https://mlatcl.github.io/advds/notes/mds.pdf`)

MDS Summary

MDS Summary

Method	Map Exp.	Map Lin.	Data Sup.	Constr. & Prop. Pres. M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	Code
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

MDS Summary

Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.	Sup.	M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

When to use MDS:

MDS Summary

Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.	Sup.	M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

When to use MDS:

- We have a distance matrix as input rather than a data matrix
- Preserve distances in reduced space
- Don't need an explcit DR mapping to run on test data
- Preserve the ambient space geometry in the reduced space

Isometric Mapping

Before Isomap... Nearest neighbors!

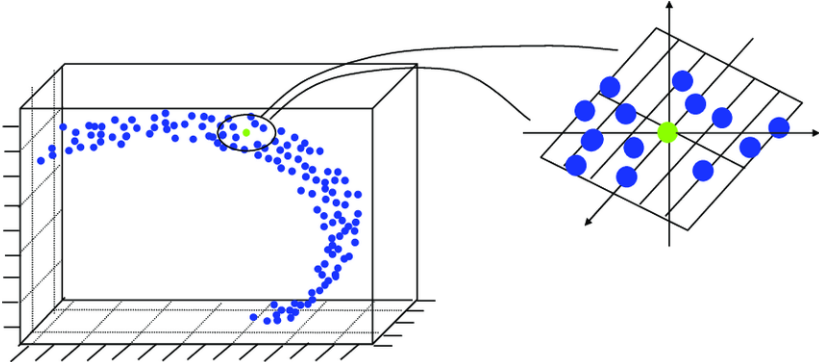
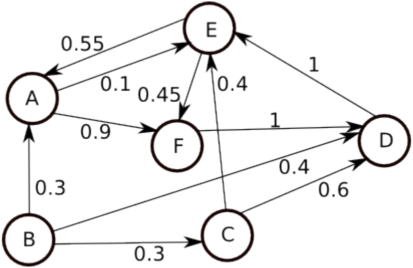


Image from: `https://www.researchgate.net/publication/35199451_Location_Estimation_in_Sensor_Networks`

Before Isomap... Graphs!

What is a graph?



A graph is collection of

- **Nodes** (e.g., A)
- **Edges** (e.g., arrow between B and A)
- **Edge weights** (e.g., 0.3)

Image from: `https://stackoverflow.com/questions/23947839/distance-between-two-nodes-in-weighted-directed-graph`

Nearest neighbor graph

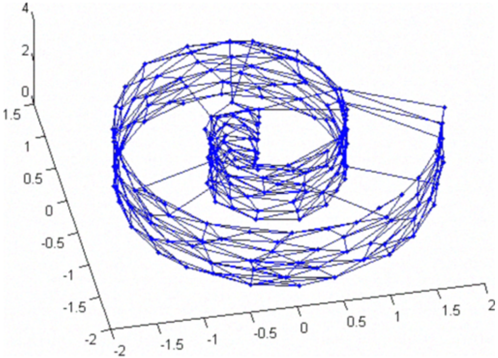


Image from: `https://www.researchgate.net/publication/221531607_A_robust_nonlinear_projection_method_using_the_neural_as_network`

What is Isomap?

Isomap (Isometric Mapping) is a nonlinear dimensionality reduction method that:

- Preserves **geodesic distances** (not just Euclidean distances)
- Extends classical MDS to nonlinear manifolds

Key idea: Approximate manifold distances with shortest paths on a neighborhood graph, then apply MDS.

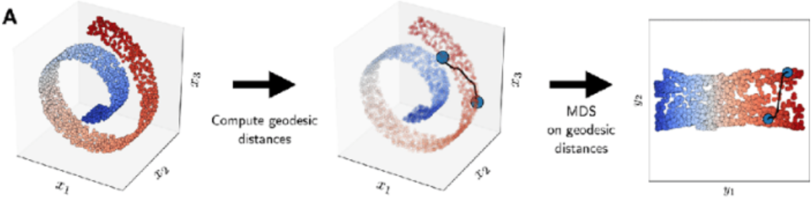


Figure: Figure borrowed from Tsai [2010].

Steps of Isomap

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Inputs: Distance, & number of nearest neighbors

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   - Edges E defined using k-nearest neighbors

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Steps of Isomap

Inputs: Distance, & number of nearest neighbors

1. Construct a graph where:
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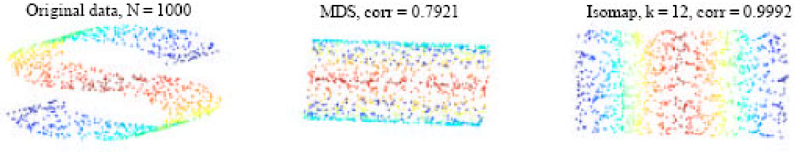
   - Edges E defined using k-nearest neighbors
2. Compute geodesic distance matrix $D^{\text{geo}} \in \mathbb{R}^{p \times p}$ using shortest path distances in G
3. Apply classical MDS to D^{geo}

Isomap vs MDS

The advantage of a nearest-neighbors approach...



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Isomap Summary

Isomap Summary

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	Exp.	Lin.	Sup.	M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

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MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

When to Use Isomap

- Your data lies on a low-dimensional **nonlinear** manifold

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When to Use Isomap

- Your data lies on a low-dimensional **nonlinear** manifold
- You want a global embedding that preserves intrinsic geometry

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MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn

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Lab 5: MDS & Isomap

Go to Lab5_IsomapMDS.ipynb

t-SNE

Stochastic Neighbor Embedding (SNE): Overview

SNE constructs a probability distribution over neighbors for each point $\mathbf{x}_i$ based on distances in high-dimensional space.

$$g(\mathbf{x}_i, \mathbf{x}) = \exp\left(-\frac{\|\mathbf{x}_i - \mathbf{x}\|_2^2}{2\sigma_i^2}\right).$$

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The conditional probability that $\mathbf{x}_i$ picks $\mathbf{x}_j$ as neighbor:

$$p_{j|i} = \frac{g(\mathbf{x}_i, \mathbf{x}_j)}{\sum_{r \neq i} g(\mathbf{x}_i, \mathbf{x}_r)}.$$

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Note: $p_{j|i} \neq p_{i|j}$ due to different σ_i values. *dirty secret... this is just a directed graph!*

Perplexity and Bandwidth Selection

Perplexity and Bandwidth Selection

Perplexity controls local scale for each point:

$$\mathrm{Perp}(P_i) = 2^{H(P_i)}, \quad H(P_i) = -\sum_j p_{j|i} \log_2 p_{j|i}.$$

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- Low perplexity $\implies$ tight local neighborhoods
- High perplexity $\implies$ broader neighborhoods

Binary search is used to find σ_i so that the perplexity matches a user-defined value.

- Small $\sigma_i \implies$ dense distribution
- Large $\sigma_i \implies$ sparse distribution

Low-dimensional Embedding

Low-dimensional Embedding

Embed points $\mathbf{z}_i \in \mathbb{R}^k$ with kernel

$$h(\mathbf{z}_i, \mathbf{z}_j) = \exp\left(-\|\mathbf{z}_i - \mathbf{z}_j\|_2^2\right),$$

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and conditional probabilities

$$q_{j|i} = \frac{h(\mathbf{z}_i, \mathbf{z}_j)}{\sum_{r \neq i} h(\mathbf{z}_i, \mathbf{z}_r)}.$$

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$$C = \sum_i \sum_j p_{j|i} \log \frac{p_{j|i}}{q_{j|i}}.$$

- $p_{j|i}$ probability that $\mathbf{x}_i$ picks $\mathbf{x}_j$ as its neighbor in ambient space
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SNE uses gradient descent on $\{\mathbf{z}_i\}$, where gradients resemble forces pulling embedded points closer or pushing them apart.

Challenges of SNE

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1. **Asymmetry**: $p_{j|i} \neq p_{i|j}$ complicates interpretation.

2. **Crowding problem**: Moderate distances in high-D collapse into tight clusters in low-D.

3. **Optimization difficulty**: KL divergence is sensitive & hard to optimize.

t-SNE: *Symmetrized* Probabilities

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- Using a **joint probability** distribution (with n samples)

$$p_{ij} = \frac{p_{i|j} + p_{j|i}}{2n}$$

t-SNE: *Symmetrized* Probabilities

t-SNE improves on SNE by:

- Using a **joint probability** distribution (with n samples)

$$p_{ij} = \frac{p_{i|j} + p_{j|i}}{2n}$$

- Modeling q_{ij} in the low-dimensional space with a **Student's-t distribution**:

$$q_{ij} = \frac{(1 + \|\mathbf{z}_i - \mathbf{z}_j\|^2)^{-1}}{\sum_{r \neq s}(1 + \|\mathbf{z}_r - \mathbf{z}_s\|^2)^{-1}}$$

t-SNE: *Fixing the Crowding Problem*

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Why use Student's-t?

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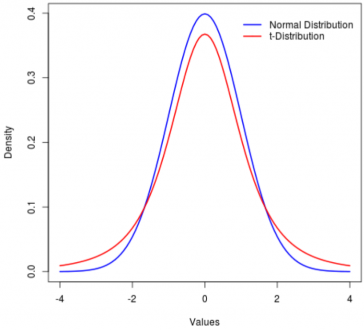
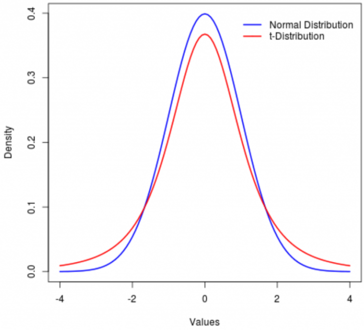


t-SNE: *Fixing the Crowding Problem*

Why use Student's-t?



The heavy tails of the t-distribution allow moderate distances to be represented more accurately — solving the **crowding problem**.

t-SNE: *Optimization*

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t-SNE minimizes the KL divergence between p_{ij} and q_{ij}:

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Gradient with respect to $\mathbf{z}_i$:

$$\frac{\partial C}{\partial \mathbf{z}_i} = 4 \sum_j (p_{ij} - q_{ij})(\mathbf{x}_i - \mathbf{z}_j)(1 + \|\mathbf{z}_i - \mathbf{z}_j\|^2)^{-1}$$

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Points repel or attract depending on similarity; repulsion is bounded (unlike in SNE) [Böhm et al., 2022]

Parameter Tuning

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t-SNE parameters:

- **Perplexity** (typical range: 5–50)
- **Learning rate**
- **Number of iterations**

Parameter Tuning

t-SNE parameters:

- **Perplexity** (typical range: 5–50)
- **Learning rate**
- **Number of iterations**

Effect of tuning: t-SNE is sensitive to parameter choices... see this website for parameter tuning intuition

t-SNE objective function is non-convex, so run multiple times and keep the lowest objective value

Summary (Part 1)

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- t-SNE improves SNE via:

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  - Symmetric probabilities
  - Heavy-tailed Student's-t distribution
  - Better optimization
- t-SNE slow ($O(n^2)$), Barnes-Hut t-SNE is faster ($O(n(\log(n)))$)

Summary (Part 2)

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Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.	Sup.	M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn

Summary (Part 2)

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LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn

When to use t-SNE

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MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
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When to use t-SNE

- t-SNE preserves local structure using probability distributions

Summary (Part 2)

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MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn

When to use t-SNE

- t-SNE preserves local structure using probability distributions
- Widely used for visualizing high-dimensional data

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PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
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MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn

When to use t-SNE

- t-SNE preserves local structure using probability distributions
- Widely used for visualizing high-dimensional data
- Often outputs *very* clustered data

Handbook for using t-SNE!

Lab 6: t-SNE

Go to Lab6_tSNE.ipynb &
`https://distill.pub/2016/misread-tsne/`

UMAP

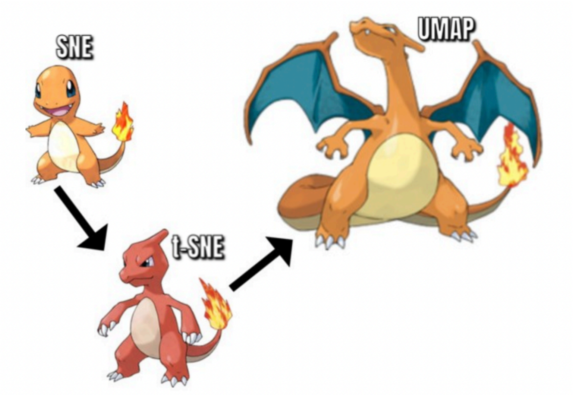

Figure: Image from https://arize.com/blog-course/reduction-of-dimensionality-top-techniques/.

Uniform Manifold Approximation (UMAP)

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Goal: preserve the topological structure of the data manifold

Inputs

Given dataset $\{\mathbf{x}_1, \ldots, \mathbf{x}_p\}$

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UMAP is based in category theory & mathematical topology.. we will sweep this foundation under the rug and consider the computational machinery instead

Algorithm Step 1: Ambient Graph Building

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3. **Define edge weights:** Edge weights from i to i_j:

$$\mathbf{w}_h(\mathbf{x}_i, \mathbf{x}_{i_j}) = \exp\left(\frac{-\max(0, d(\mathbf{x}_i, \mathbf{x}_{i_j}) - \rho_i)}{\sigma_i}\right)$$

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Interpretation: edge weights are probabilities that a 1-simplex exists; combined weights: probability that at least one edge exists.

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Edge weights between $\mathbf{z}_i$ and $\mathbf{z}_j$ in reduced space are

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Where a and b are hyperparameters chosen by a non-linear least squares fit to

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Default parameters: $a = 1.577$, $b = 0.8951$

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$$C = \sum_{e} w_h(e) \log\left(\frac{w_h(e)}{w_l(e)}\right) + (1 - w_h(e)) \log\left(\frac{1 - w_h(e)}{1 - w_l(e)}\right)$$

Parameter Tuning & Resources

Key Parameters:

- **Number of neighbors:** balance local vs. global structure
- **Minimum distance:** desired separation of points in latent space

Visualizing parameter tuning and comparison to t-SNE

Links & Resources:

- UMAP: Uniform Manifold Approximation and Projection
- UMAP Documentation
- Understanding UMAP
- Inverse transform
- Aligned UMAP

UMAP Summary (Part 1)

- Preserves global and local structure

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Method	Map		Data	Constr. & Prop. Pres.							Code
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PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
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MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn
UMAP	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	github

Link for a tutorial on how to use UMAP.

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Link for a tutorial on how to use UMAP.

***We can make implicit methods explicit** by optimizing for function parameters for embeddings instead of embedding coordinates!!! Damrich et al. [2022]*

Lab 7: UMAP

Go to Lab7_UMAP.ipynb &
`https://pair-code.github.io/understanding-umap/`

Autoencoders

Outline

1. Vanilla Autoencoders
2. Denoising Autoencoders
3. Variational Autoencoders
4. Convolutional Autoencoders
5. Graph Autoencoders

Neural Networks

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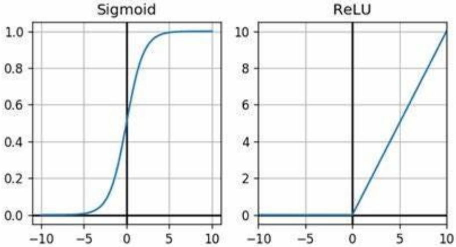

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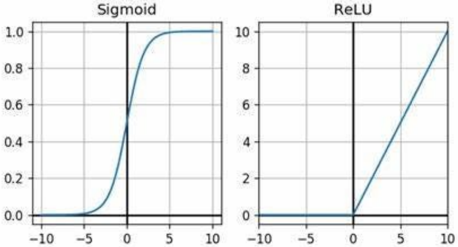

The "neural network function" maps

$$\mathbf{x} \mapsto \sigma(\mathbf{W}_\ell \sigma(\mathbf{W}_{\ell-1} \cdots \sigma(\mathbf{W}_1(\mathbf{x})) \cdots))$$

Vanilla Autoencoder

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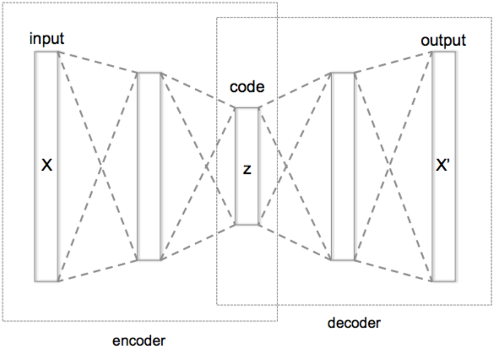

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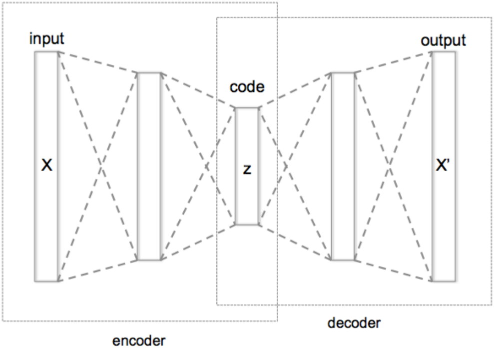

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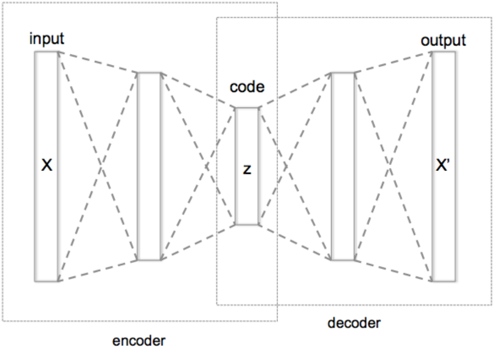

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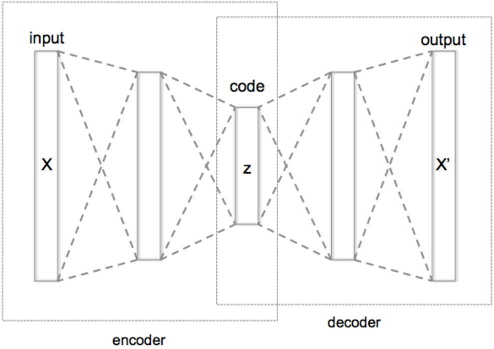

- ϕ: Encoder compresses the input into a lower-D representation
- ψ: Decoder reconstructs the input from the lower-D representation
- Loss function: Typically Mean Squared Error (MSE) between the input and the reconstructed output

Autoencoders vs PCA

Autoencoders optimize reconstruction error

$$\arg\min_{\mathrm{Enc},\mathrm{Dec}} \sum_{i=1}^{n} L(\mathbf{x}_i, \mathrm{Dec} \circ \mathrm{Enc}(\mathbf{x}_i)).$$

- Define $\mathrm{Enc}(\mathbf{x}) = \mathbf{A}^\top \mathbf{x}$ and $\mathrm{Dec}(\mathbf{z}) = \mathbf{A}\mathbf{z}$.
- Take the mean squared error (MSE) loss $L(\mathbf{x}, \hat{\mathbf{x}}) = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2$.

$$\arg\min_{\mathbf{A}^\top \mathbf{A} = \mathbf{I}} \sum_{i=1}^{n} \|\mathbf{x}_i - \mathbf{A}\mathbf{A}^\top \mathbf{x}_i\|_2^2.$$

- This is *exactly* PCA

But, there's a deeper connection... see Rolinek et al. [2019]; Bao et al. [2020].

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Problem: Marginal likelihood is intractable because we don't know the posterior

$$p_\theta(\mathbf{x}) = \int_{\mathbf{z}} p_\theta(\mathbf{x}, \mathbf{z}) d\mathbf{z} = \int_{\mathbf{z}} p_\theta(\mathbf{z}) p_\theta(\mathbf{x}|\mathbf{z}) d\mathbf{z}$$

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Although $p_\theta(\mathbf{z}|\mathbf{x})$ is also intractible... we can estimate it!

Estimate $p_\theta(\mathbf{z}|\mathbf{x})$

Goal: approximate $p = p_\theta(\mathbf{z}|\mathbf{x})$ with $q = q_\phi(\mathbf{z}|\mathbf{x})$.
KL Divergence

$$
\begin{aligned}
\mathrm{D}_{\mathrm{KL}}(q||p) &= \int_{\mathbf{z}} q \log(q/p) d\mathbf{z} && (7)\\
&= \mathbb{E}_q \left[\log(q/p)\right] && (8)\\
&= \mathbb{E}_q[\log q] - \mathbb{E}_q[\log p] && (9)\\
&= \mathbb{E}_q[\log q] - \mathbb{E}_q[\log(p_\theta(\mathbf{z},\mathbf{x})/p_\theta(\mathbf{x}))] && (10)\\
&= \mathbb{E}_q[\log q] - \mathbb{E}_q[\log p_\theta(\mathbf{z},\mathbf{x})] + \mathbb{E}_q[\log p_\theta(\mathbf{x})] && (11)\\
&= \log p_\theta(\mathbf{x}) - \mathbb{E}_q[\log p_\theta(\mathbf{z},\mathbf{x}) - \log q_\phi(\mathbf{z}|\mathbf{x})] && (12)
\end{aligned}
$$

The expected information loss when we model p with q.

Problem: $\log p_\theta(\mathbf{x})$ intractible

More with KL Divergence

From before, we have

$$\log p_\theta(x) = \mathrm{D}_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}|\mathbf{x})) + \mathbb{E}_q[\log p_\theta(\mathbf{z}, \mathbf{x}) - \log q_\phi(\mathbf{z}|\mathbf{x})]$$

Notice $\mathrm{D}_{\mathrm{KL}}(q||p) > 0$

Enter the Evidence Lower BOund (ELBO)!

$$\log p_\theta(x) \geq \mathbb{E}_q[\log p_\theta(\mathbf{z}, \mathbf{x}) - \log q_\phi(\mathbf{z}|\mathbf{x})] = \mathrm{ELBO}$$

The Glory of ELBO

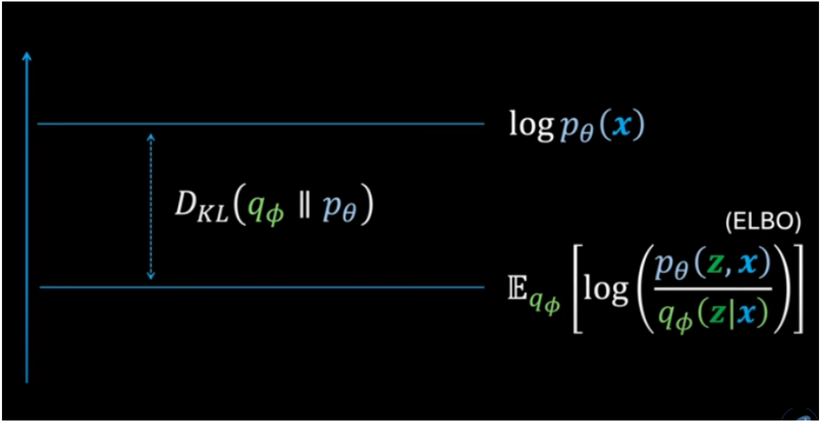

Figure: Image borrowed from `https://www.youtube.com/watch?v=HBYQvKlaE0A`.

Maximizing ELBO maximizes $p_\theta(\mathbf{x})$ and $\mathrm{D}_{\mathrm{KL}}(q||p)$!

ELBO Optimization

Stochastic gradient descent on ELBO... but issues with estimation of gradient...

Reparameterization Trick:

- $q_\phi(\mathbf{z}|\mathbf{x}) = g(\phi, \mathbf{x}, \epsilon)$
- $\epsilon \sim p(\epsilon)$

Now expectation is taken over $p(\epsilon)$, rather than q

$$\mathcal{L}(\mathbf{x}) = \mathbb{E}_{p(\epsilon)}[\log p_\theta(\mathbf{z}, \mathbf{x}) - \log q_\phi(\mathbf{z}|\mathbf{x})]$$

Exercise: we approximate ELBO with L samples using

$$\mathrm{ELBO} \approx \frac{1}{L} \sum_{i=1}^{L} \log p_\theta(\mathbf{x}|\mathbf{z}) - \mathrm{D}_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x}) || p_\theta(\mathbf{z}))$$

Latent Space Simplifying Assumptions

From before ... the VAE is trained to maximize

$$\mathrm{ELBO} \approx \frac{1}{L} \sum_{i=1}^{L} \log p_\theta(\mathbf{x}|\mathbf{z}) - \mathrm{D}_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}))$$

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- Prior: $p(\mathbf{z}) = \mathcal{N}(0, I)$
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Exercise: Show that this loss becomes

$$\mathrm{D}_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z})) = \frac{-1}{2}[\log \boldsymbol{\sigma}^2 - \boldsymbol{\mu}^2 - \boldsymbol{\sigma}^2 + 1].$$

VAE Architecture

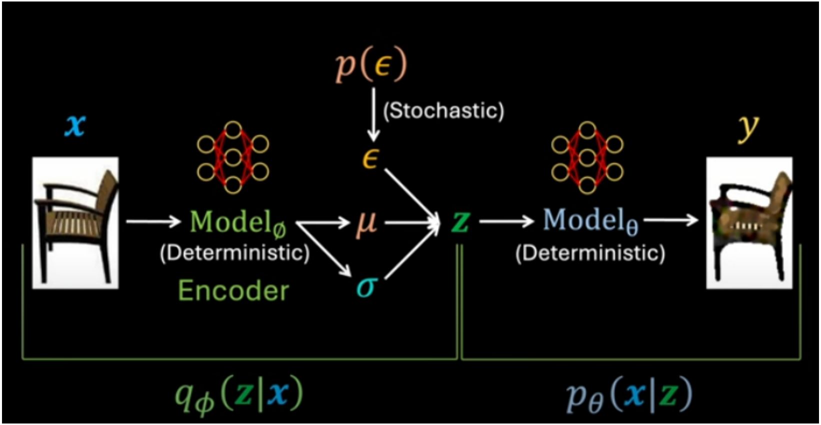

Figure: Image borrowed from
`https://www.youtube.com/watch?v=HBYQvKlaE0A`.

The VAE Loss

$$-\mathrm{ELBO} \approx -\frac{1}{L}\sum_{i=1}^{L} \log p_\theta(\mathbf{x}|\mathbf{z}) + \mathrm{D}_{\mathrm{KL}}(q_\phi(\mathbf{z}|\mathbf{x})||p_\theta(\mathbf{z}))$$

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"Reconstruction error" $\log p_\theta(\mathbf{x}|\mathbf{z})$

- Gaussian assumption: $(\mathbf{x} - \hat{\mathbf{x}})^2$ *(takes work to show)*
- Bernoulli assumption: $\mathrm{BCE}(\hat{\mathbf{x}}, \mathbf{x})$

Similarity between predicted $q_\phi(\mathbf{z}|\mathbf{x})$ & true $p_\theta(\mathbf{z})$) latent distributions

We assume $p_\theta(\mathbf{z})$) is Gaussian: $p_\theta(\mathbf{z}) = \mathcal{N}(0, I)$

VAE Summary

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When to use VAEs: generative modeling and denoising

VAE Summary

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Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.	Sup.	M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn
UMAP	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	github
DAE	✓	✗	✗	✗	✓	✓	✗	✗	✗	✗	github
VAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✗	github

Convolutional Autoencoder

Continuous Convolution

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Consider functions: $f, g : \mathbb{R} \to \mathbb{R}$.

$$f * g(x) = \int_u f(u)g(x-u)du$$

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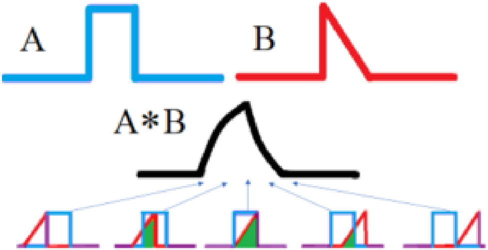

Figure:
https://www.statisticshowto.com/convolution-integral-simple-definition/

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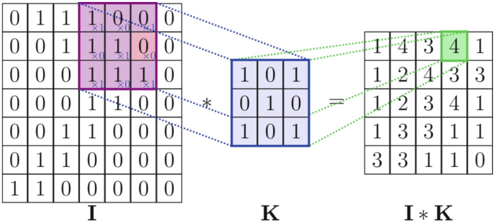

Figure: `https://www.researchgate.net/figure/A-visualization-of-the-discrete-convolution-operation-in-2D_fig5_372609466`

Sliding convolutional filter.

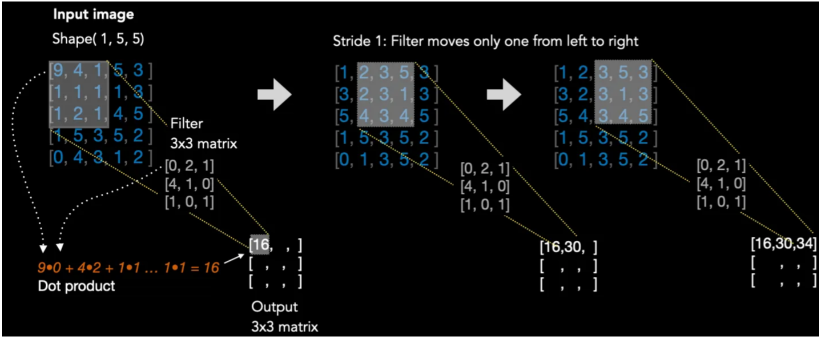

Figure:
https://medium.com/advanced-deep-learning/cnn-operation-with-2-kernels
-resulting-in-2-feature-mapsunderstanding-the-convolutional-filter-c4aad26cf32

Certain filters extract different patterns.

Input image

Convolution Kernel

$$\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

Feature map

Figure: `https://timdettmers.com/2015/03/26/convolution-deep-learning/`

Convolutional Autoencoders

Encoder and decoder are convolutional filters

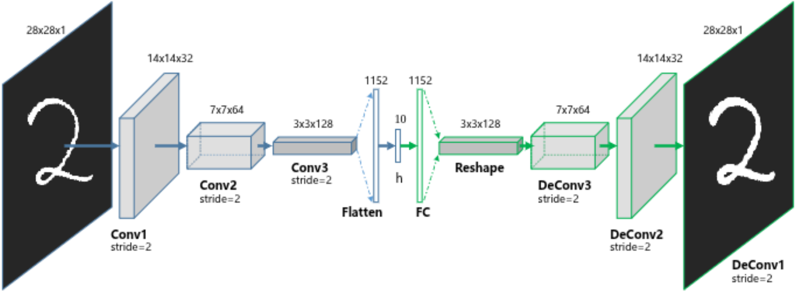

Figure: [Guo et al., 2017]

CAE Summary

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Graph Convolutional Autoencoder

Convolutions and Fourier Transform

Convolutions and Fourier Transform

Let $\mathcal{F}$ be the Fourier transform. Consider functions: $f, g : \mathbb{R} \to \mathbb{R}$. Then, under certain assumptions

$$\mathcal{F}(f * g) = \mathcal{F}(f)\mathcal{F}(g)$$

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- Multiplication is easy!
- Decreases the computational cost of convolution

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$$\mathbf{V} = \frac{1}{\sqrt{N}} \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 1 & \omega & \cdots & \omega^{d-1} \\ 1 & \omega^2 & \cdots & \omega^{2(d-1)} \\ \vdots & \vdots & \ddots & \vdots \\ 1 & \omega^{p-1} & \cdots & \omega^{(d-1)(d-1)} \end{bmatrix}.$$

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- The DFT of a timeseries $\mathbf{x} = [x(1)|x(2)|\cdots|x(d)]^\top$ is $\mathbf{Vx}$
- Transforms from time to frequency domain
- Can also be used on other signals (features other than time)

Graph Data

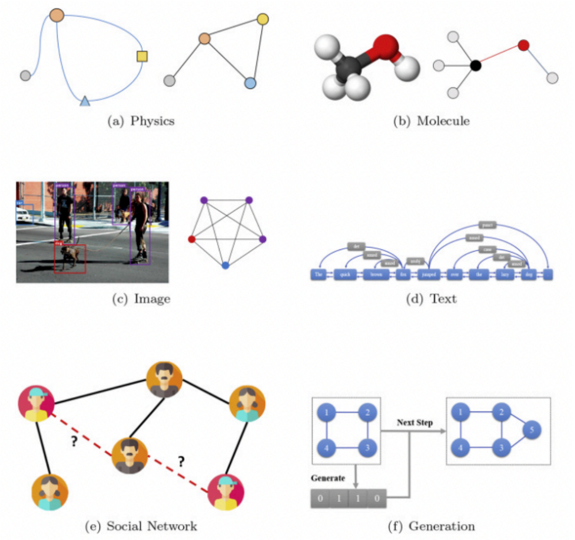
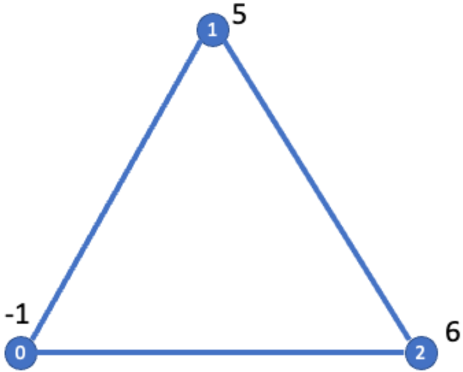


Figure: [Zhou et al., 2020]

Graph Signals

A graph signal is a vector with a value at every node on the graph.



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On the cycle graph, the GFT reduces to the DFT.

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$$\mathcal{F}(\mathbf{x} * \mathbf{g}) = \mathcal{F}(\mathbf{x}) \cdot \mathcal{F}(\mathbf{g}) = \mathbf{\Lambda}_{\mathbf{U}^\top \mathbf{g}} \mathbf{U}^\top \mathbf{x}$$

Where $\cdot$ is point-wise multiplication and

$$\mathbf{\Lambda}_g = \begin{bmatrix} (\mathbf{U}^\top \mathbf{g})_1 & \cdots & 0 \\ \vdots & \ddots & \vdots \\ 0 & \cdots & (\mathbf{U}^\top \mathbf{g})_d \end{bmatrix}$$

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See Kipf and Welling [2016b] for details.

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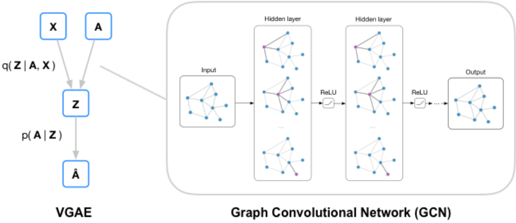

Figure: Code and image from `https://github.com/tkipf/gae`

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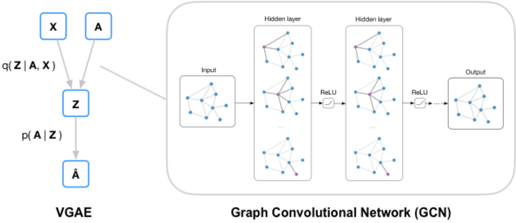

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Further reading: graph-based architectures [Wu et al., 2020] and generalizations of graph autoencoders [Hajij et al., 2021]

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DR for Visualization

Notes from a CVPR Tutorial

Practical Tips for DR Visualization

Do's:

- Each DR method has a **scale parameter** (e.g., kernel bandwidth, perplexity) — explore its effect.
- Be creative with **inputs** (e.g., similarity, probability, or relational data).
- Be creative with **outputs** (e.g., use low-D coords as RGB values or overlays).
- Look for **patterns, clusters, and relations**.
- Use DR to **generate hypotheses**, not to prove them.
- Explore time–series visualization using DR is an emerging research direction.

Don'ts:

- Avoid "**proof by visualization.**" DR plots are for exploration, not evidence. Don't draw firm conclusions solely from DR plots.
- Low-dimensional metric spaces cannot represent all non-metric similarities (real-world data often violates the triangle inequality).

Summary

- *Linear algebra review*
- *Taxonomy of DR*
- *Linear DR:* PCA & LDA
- *Nonlinear DR:* MDS, Isomap, t-SNE, & UMAP
- *Autoencoders:* DAE, VAE, CAE, VGAE

The end, thank you!

Special thanks to Gustau Camps-Valls, Kai Hendrik-Cors, Homer Durand, Vassilis Sitokonstantinou & Tristan Williams

Extra Topics

Extra Topics: Outline

1. Gradient Descent
2. Dynamic Mode Decomposition [Tu, 2013]
3. Koopman Mode Decomposition [Brunton et al., 2021]

Gradient Descent

Gradient Descent: Overview

- Gradient Descent is an optimization algorithm to minimize a function

- It iteratively moves in the direction of steepest descent (negative gradient)

Objective

Minimize a loss function $f(\theta)$, where:

- θ are the model parameters

Update Rule

Update Rule

At each iteration, the parameters are updated as:

$$\theta_{t+1} = \theta_t - \eta \nabla f(\theta_t)$$

- θ_t: Current parameters at iteration t
- η: Learning rate (step size)
- $\nabla f(\theta_t)$: Gradient of the loss function

Dynamic Mode Decomposition

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- Goal: Decompose the system into state space and temporal patterns, thus reducing temporal and state space dimensions.

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Linear Systems

If $f(\mathbf{x}) = \mathbf{A}\mathbf{x}$, the system is linear:

$$\mathbf{x}(t+\tau) = \mathbf{A}\mathbf{x}(t)$$

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Linear Systems
If $f(\mathbf{x}) = \mathbf{Ax}$, the system is linear:

$$\mathbf{x}(t+\tau) = \mathbf{Ax}(t)$$

$\Rightarrow$ Solutions evolve through powers of A: eigenvalues/eigenvectors govern behavior.

Dynamic Mode Decomposition

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Assume that the data is sampled from the timeseries

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- $\phi_j \in \mathbb{C}^n$ **feature patterns** (dynamic modes)
- $\omega_j \in \mathbb{C}$ **temporal characteristics** (continuous time eigenvalues)
- $b_j \in \mathbb{R}$ scalar loadings (a.k.a. amplitudes)

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DMD analyzes the stability of this system and generates spatial patterns in $\mathbb{R}^n$ (ambient space)

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Solve via variable projection method.

Further Reading

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- Multiverse of DMD [Colbrook, 2023]
- Physics-informed DMD Baddoo et al. [2023]
- Generalizing DMD: Modern Koopman theory [Brunton et al., 2021]

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Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.	Sup.	M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn
UMAP	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	github
D & CAE	✓	✗	✗	✗	✓	✓	✗	✗	✗	✗	github
VAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✗	github
VGAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✓	github
DMD	✓	✓	✗	✓	✗	✓	✗	✗	✗	✗	github

Bonus lab!

Tutorial 1 here.

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Problem: there's no linear map that goes from $\mathbf{x}(t)$ to $\mathbf{x}(t+\tau)$...

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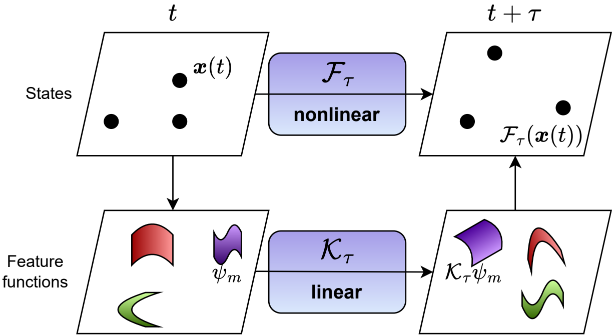

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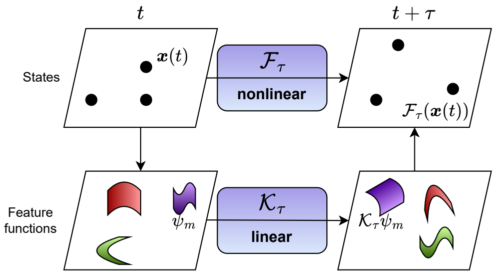

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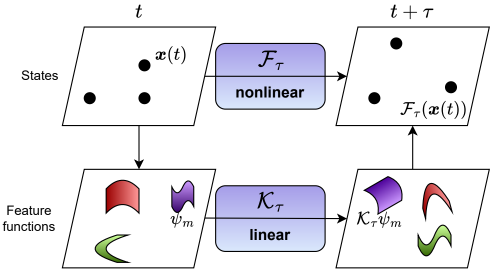

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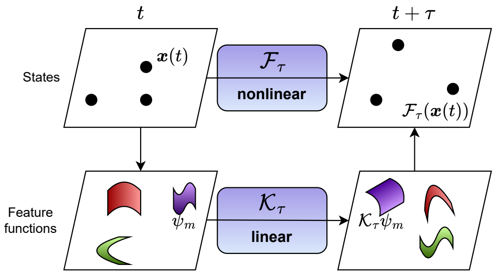

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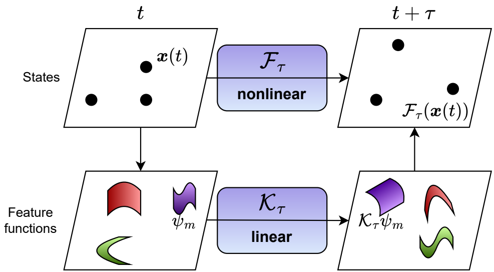

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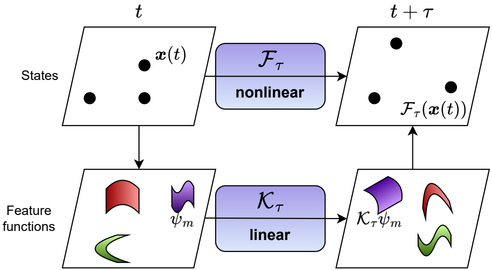

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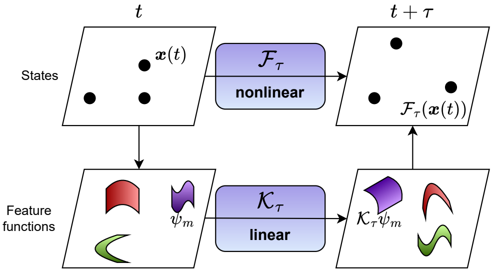

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Name	Definition
Eigenfunctions of $\mathbf{X}$	$\boldsymbol{\Phi} = \mathbf{K_X}\,\mathbf{Q}\,\boldsymbol{\Sigma}^\dagger\,\mathbf{W}$
Koopman modes	$\mathbf{B} = \mathbf{X}\mathbf{Q}^T\boldsymbol{\Sigma}^\dagger\mathbf{W}^{-1}$

KMD Summary

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When to use KMD: nonlinear timeseries DR

Method	Map		Data	Constr. & Prop. Pres.							Code
	Exp.	Lin.	Sup.	M.F.	A.E.	Rec.	Var.	Dist.	Geo.	Top	
PCA	✓	✓	✗	✓	✗	✓	✓	✗	✗	✗	sklearn
LDA	✓	✓	✓	✓	✗	✓	✓	✗	✗	✗	sklearn
MDS	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
Isomap	✗	✗	✗	✗	✗	✗	✗	✗	✓	✗	sklearn
t-SNE	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	sklearn
UMAP	✗	✗	✗	✗	✗	✗	✗	✓	✗	✓	github
D & CAE	✓	✗	✗	✗	✓	✓	✗	✗	✗	✗	github
VAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✗	github
VGAE	✓	✗	✗	✗	✓	✓	✗	✓	✗	✓	github
DMD	✓	✓	✗	✓	✗	✓	✗	✗	✗	✗	github
KMD	✗	✓	✗	✓	✗	✓	✗	✗	✗	✗	github

Bonus lab!

Tutorials here:

- Extended (Kernel) DMD
- Koopman Mode Decomposition

References I

Travis Askham and J Nathan Kutz. Variable projection methods for an optimized dynamic mode decomposition. SIAM Journal on Applied Dynamical Systems, 17(1):380–416, 2018.

Peter J Baddoo, Benjamin Herrmann, Beverley J McKeon, J Nathan Kutz, and Steven L Brunton. Physics-informed dynamic mode decomposition. Proceedings of the Royal Society A, 479(2271): 20220576, 2023.

Mukund Balasubramanian and Eric L Schwartz. The isomap algorithm and topological stability. Science, 295(5552):7–7, 2002.

Xuchan Bao, James Lucas, Sushant Sachdeva, and Roger B Grosse. Regularized linear autoencoders recover the principal components, eventually. Advances in Neural Information Processing Systems, 33: 6971–6981, 2020.

Mikhail Belkin and Partha Niyogi. Laplacian eigenmaps and spectral techniques for embedding and clustering. Advances in neural information processing systems, 14, 2001.

References II

Jan Niklas Böhm, Philipp Berens, and Dmitry Kobak. Attraction-repulsion spectrum in neighbor embeddings. Journal of Machine Learning Research, 23(95):1–32, 2022.

Steven L Brunton, Marko Budišić, Eurika Kaiser, and J Nathan Kutz. Modern koopman theory for dynamical systems. arXiv preprint arXiv:2102.12086, 2021.

Matthew J Colbrook. The multiverse of dynamic mode decomposition algorithms. arXiv preprint arXiv:2312.00137, 2023.

Sebastian Damrich, Jan Niklas Böhm, Fred A Hamprecht, and Dmitry Kobak. From *t*-SNE to UMAP with contrastive learning. arXiv preprint arXiv:2206.01816, 2022.

Scott TM Dawson, Maziar S Hemati, Matthew O Williams, and Clarence W Rowley. Characterizing and correcting for the effect of sensor noise in the dynamic mode decomposition. Experiments in Fluids, 57:1–19, 2016.

References III

P Thomas Fletcher, Conglin Lu, Stephen M Pizer, and Sarang Joshi. Principal geodesic analysis for the study of nonlinear statistics of shape. IEEE transactions on medical imaging, 23(8):995–1005, 2004.

Matan Gavish and David L Donoho. The optimal hard threshold for singular values is $4/\sqrt{3}$. IEEE Transactions on Information Theory, 60 (8):5040–5053, 2014.

Xifeng Guo, Xinwang Liu, En Zhu, and Jianping Yin. Deep clustering with convolutional autoencoders. In Neural Information Processing: 24th International Conference, ICONIP 2017, Guangzhou, China, November 14-18, 2017, Proceedings, Part II 24, pages 373–382. Springer, 2017.

Mustafa Hajij, Ghada Zamzmi, Theodore Papamarkou, Vasileios Maroulas, and Xuanting Cai. Simplicial complex representation learning. arXiv preprint arXiv:2103.04046, 2021.

Harold Hotelling. Analysis of a complex of statistical variables into principal components. Journal of educational psychology, 24(6):417, 1933.

References IV

Thomas N Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. arXiv preprint arXiv:1609.02907, 2016a.

Thomas N Kipf and Max Welling. Variational graph auto-encoders. arXiv preprint arXiv:1611.07308, 2016b.

Bernard O. Koopman. Hamiltonian systems and transformation in Hilbert space. Proceedings of the National Academy of Sciences, 17(5): 315–318, 1931.

Kenneth Kreutz-Delgado, Joseph F Murray, Bhaskar D Rao, Kjersti Engan, Te-Won Lee, and Terrence J Sejnowski. Dictionary learning algorithms for sparse representation. Neural computation, 15(2): 349–396, 2003.

Joseph B Kruskal and Myron Wish. Multidimensional scaling. Number 11. Sage, 1978.

John A Lee and Michel Verleysen. Nonlinear dimensionality reduction. Springer Science & Business Media, 2007.

Gilad Lerman and Tyler Maunu. An overview of robust subspace recovery. Proceedings of the IEEE, 106(8):1380–1410, 2018.

References V

Miguel D Mahecha, Fabian Gans, Gunnar Brandt, Rune Christiansen, Sarah E Cornell, Normann Fomferra, Guido Kraemer, Jonas Peters, Paul Bodesheim, Gustau Camps-Valls, et al. Earth system data cubes unravel global multivariate dynamics. Earth System Dynamics, 11(1): 201–234, 2020.

Nathan Mankovich, Gustau Camps-Valls, and Tolga Birdal. Fun with flags: Robust principal directions via flag manifolds. In Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition, pages 330–340, 2024.

Leland McInnes, John Healy, and James Melville. UMAP: Uniform manifold approximation and projection for dimension reduction. arXiv preprint arXiv:1802.03426, 2018.

Igor Mezić. Analysis of fluid flows via spectral properties of the koopman operator. Annual Review of Fluid Mechanics, 45:357–378, 2013.

Xavier Pennec. Barycentric subspace analysis on manifolds. The Annals of Statistics, 2018.

References VI

Michal Rolinek, Dominik Zietlow, and Georg Martius. Variational autoencoders pursue PCA directions (by accident). In IEEE Conference on Computer Vision and Pattern Recognition, pages 12406–12415, 2019.

Clarence W. Rowley, Igor Mezić, Shervin Bagheri, Philipp Schlatter, and Dan S. Henningson. Spectral analysis of nonlinear flows. Journal of Fluid Mechanics, 641:115–127, 2009.

Peter J Schmid. Dynamic mode decomposition of numerical and experimental data. Journal of fluid mechanics, 656:5–28, 2010.

Jonathon Shlens. A tutorial on principal component analysis. arXiv preprint arXiv:1404.1100, 2014.

Gilbert Strang. Linear algebra and its applications. 2000.

Tom Szwagier and Xavier Pennec. The curse of isotropy: From principal components to principal subspaces. 2024. URL https://arxiv.org/abs/2307.15348.

Tom Szwagier and Xavier Pennec. Nested subspace learning with flags. arXiv preprint arXiv:2502.06022, 2025.

References VII

Alaa Tharwat, Tarek Gaber, Abdelhameed Ibrahim, and Aboul Ella Hassanien. Linear discriminant analysis: A detailed tutorial. AI communications, 30(2):169–190, 2017.

FS Tsai. Comparative study of dimensionality reduction techniques for data visualization. Journal of artificial intelligence, 3(3):119–134, 2010.

Jonathan H Tu. Dynamic mode decomposition: Theory and applications. PhD thesis, Princeton University, 2013.

Laurens Van der Maaten and Geoffrey Hinton. Visualizing data using t-sne. Journal of machine learning research, 9(11), 2008.

Pascal Vincent, Hugo Larochelle, Yoshua Bengio, and Pierre-Antoine Manzagol. Extracting and composing robust features with denoising autoencoders. In Proceedings of the 25th international conference on Machine learning, pages 1096–1103, 2008.

Matthew O Williams, Ioannis G Kevrekidis, and Clarence W Rowley. A data–driven approximation of the Koopman operator: Extending dynamic mode decomposition. Journal of Nonlinear Science, 25: 1307–1346, 2015.

Zonghan Wu, Shirui Pan, Fengwen Chen, Guodong Long, Chengqi Zhang, and Philip S Yu. A comprehensive survey on graph neural networks. IEEE transactions on neural networks and learning systems, 32(1):4–24, 2020.

Jie Zhou, Ganqu Cui, Shengding Hu, Zhengyan Zhang, Cheng Yang, Zhiyuan Liu, Lifeng Wang, Changcheng Li, and Maosong Sun. Graph neural networks: A review of methods and applications. AI open, 1: 57–81, 2020.